

Skill Week - 7

1. To Parse Input Redirection, Open Files, Redirect stdin, Handle Missing Files, Restore Streams, Test File Inputs

```
shivam@MSI123: ~
GNU nano 8.7.1 src/input_redirect.c *
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <string.h>

int main() {
    char input[100];
    int fd, saved_stdin;

    printf("Enter input file name: ");
    scanf("%99s", input);

    fd = open(input, O_RDONLY);

    if (fd < 0) {
        perror("Error opening input file");
        return 1;
    }

    saved_stdin = dup(STDIN_FILENO);

    dup2(fd, STDIN_FILENO);
    close(fd);

    printf("Reading redirected input:\n");

    char buffer[100];
    ssize_t n = read(STDIN_FILENO, buffer, sizeof(buffer) - 1);

    if (n > 0) {
        buffer[n] = '\0';
        printf("%s", buffer);
    }

    dup2(saved_stdin, STDIN_FILENO);
    close(saved_stdin);

    printf("\nInput stream restored.\n");

    return 0;
}
```

Output:

```
shivam@MSI123:~$ gcc src/input_redirect.c -o bin/input_redirect
shivam@MSI123:~$ ./bin/input_redirect
Enter input file name: input.txt
Reading redirected input:
Hello from input file!

Input stream restored.
shivam@MSI123:~$ ./bin/input_redirect
Enter input file name: abc.txt
Error opening input file: No such file or directory
shivam@MSI123:~$ |
```

2.To Parse Output Redirection, Create Output Files, Redirect stdout, Handle Permissions, Verify Output Data, Test Redirection

```
shivam@MSI123: ~  
GNU nano 8.7.1 src/output_redirect.c *  
#include <stdio.h>  
#include <fcntl.h>  
#include <unistd.h>  
  
int main() {  
    char filename[100];  
    int fd, saved_stdout;  
  
    printf("Enter output file name: ");  
    scanf("%99s", filename);  
  
    fd = open(filename, O_WRONLY | O_CREAT | O_TRUNC, 0644);  
  
    if (fd < 0) {  
        perror("Error opening output file");  
        return 1;  
    }  
  
    saved_stdout = dup(STDOUT_FILENO);  
  
    dup2(fd, STDOUT_FILENO);  
    close(fd);  
  
    printf("This text is redirected to the output file.\n");  
    printf("Output redirection successful.\n");  
  
    fflush(stdout);  
  
    dup2(saved_stdout, STDOUT_FILENO);  
    close(saved_stdout);  
  
    printf("Output stream restored.\n");  
  
    return 0;  
}
```

Output:

```
shivam@MSI123:~$ nano src/output_redirect.c  
shivam@MSI123:~$ gcc src/output_redirect.c -o bin/output_redirect  
shivam@MSI123:~$ ./bin/output_redirect  
Enter output file name: output.txt  
Output stream restored.  
shivam@MSI123:~$ cat output.txt  
This text is redirected to the output file.  
Output redirection successful.  
shivam@MSI123:~$ ls -l output.txt  
-rw-r--r-- 1 priyanka priyanka 75 Sep 18 07:03 output.txt
```

3.To Parse Append Operator, Open Files in Append Mode, Redirect Streams, Preserve Existing Data, Validate Appends, Test Scenarios

```
shivam@MSI123: ~  
GNU nano 8.7.1 src/append_redirect.c *  
#include <stdio.h>  
#include <fcntl.h>  
#include <unistd.h>  
#include <string.h>  
  
int main() {  
    char filename[100];  
    int fd, saved_stdout;  
    char *text = "New line appended to the file.\n";  
  
    printf("Enter output file name: ");  
    scanf("%99s", filename);  
  
    fd = open(filename, O_WRONLY | O_CREAT | O_APPEND, 0644);  
  
    if (fd < 0) {  
        perror("Error opening file");  
        return 1;  
    }  
  
    saved_stdout = dup(STDOUT_FILENO);  
  
    dup2(fd, STDOUT_FILENO);  
    close(fd);  
  
    printf("%s", text);  
    fflush(stdout);  
  
    dup2(saved_stdout, STDOUT_FILENO);  
    close(saved_stdout);  
  
    printf("Data appended successfully.\n");  
  
    return 0;  
}
```

Output:

```
shivam@MSI123:~$ cd ~/OS_Skill/Skill-Week-7  
nano src/append_redirect.c  
shivam@MSI123:~$ gcc src/append_redirect.c -o bin/append_redirect  
shivam@MSI123:~$ echo "Original data" > append.txt  
shivam@MSI123:~$ cat append.txt  
Original data  
shivam@MSI123:~$ ./bin/append_redirect  
Enter output file name: append.txt  
Data appended successfully.  
shivam@MSI123:~$ cat append.txt  
Original data  
New line appended to the file.  
shivam@MSI123:~$ ./bin/append_redirect  
Enter output file name: append.txt  
Data appended successfully.  
shivam@MSI123:~$ cat append.txt  
Original data  
New line appended to the file.  
New line appended to the file.
```

4. To Parse stderr Redirection, Redirect Error Streams, Capture Error Output, Verify File Contents, Handle Failures, Test Error Cases

```
shivam@MSI123: ~  
GNU nano 8.7.1 src/stderr_redirect.c *  
#include <stdio.h>  
#include <fcntl.h>  
#include <unistd.h>  
  
int main() {  
    int fd, saved_stderr;  
  
    fd = open("error.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);  
  
    if (fd < 0) {  
        perror("Cannot open error.txt");  
        return 1;  
    }  
  
    saved_stderr = dup(STDERR_FILENO);  
  
    dup2(fd, STDERR_FILENO);  
    close(fd);  
  
    FILE *file = fopen("missing.txt", "r");  
  
    if (file == NULL) {  
        perror("Error opening missing.txt");  
    } else {  
        fclose(file);  
    }  
  
    fflush(stderr);  
  
    dup2(saved_stderr, STDERR_FILENO);  
    close(saved_stderr);  
  
    printf("stderr restored.\n");  
  
    return 0;  
}
```

Output:

```
shivam@MSI123:~$ nano src/stderr_redirect.c  
shivam@MSI123:~$ gcc src/stderr_redirect.c -o bin/stderr_redirect  
shivam@MSI123:~$ ./bin/stderr_redirect  
stderr restored.  
shivam@MSI123:~$ cat error.txt  
Error opening missing.txt: No such file or directory
```